



Prepare 50% Faster

Percentage

Assignment with Solutions

Percentage

Percentage is a method of expressing fraction or parts of any quantity into an equivalent form. Percentages are expressed in terms of hundredths i.e., the term per cent means for every hundred. A fraction whose denominator is 100 is called a percentage and the numerator of such a fraction is called the rate per cent. It is denoted by the symbol %.

$$\text{➤ } \cdot \quad \% \text{ Increase} = \frac{\text{Increase in value}}{\text{Original value}} \times 100$$

$$\text{➤ } \cdot \quad \% \text{ Decrease} = \frac{\text{Decrease in value}}{\text{Original value}} \times 100$$

$$\text{➤ } \cdot \quad \% \text{ Increase} = \left(\frac{r}{100+r} \right) \times 100$$

Where, r = decrease %

$$\text{➤ } \cdot \quad \% \text{ decrease} = \left(\frac{r}{100-r} \times 100 \right) \%$$

Where, r = increase %

$$\text{➤ } 1 = 100\%$$

$$\text{➤ } \frac{1}{2} = 50\%$$

$$\text{➤ } \frac{1}{3} = 33\frac{1}{3}\% = 33.33\%$$

$$\text{➤ } \frac{1}{4} = 25\%$$

$$\text{➤ } \frac{1}{5} = 20\%$$

$$\text{➤ } \frac{1}{6} = 16\frac{2}{3}\% = 16.66\%$$

$$\text{➤ } \frac{1}{7} = 14\frac{2}{7}\% = 14.28\%$$

$$\text{➤ } \frac{1}{8} = 12\frac{1}{2}\% = 12.50\%$$

$$\text{➤ } \frac{1}{9} = 11\frac{1}{9}\% = 11.11\%$$

$$\text{➤ } \frac{1}{10} = 10\%$$

$$\text{➤ } \frac{1}{11} = 9\frac{1}{11}\% = 9.09\%$$

$$\text{➤ } \frac{1}{12} = 8\frac{1}{3}\% = 8.33\%$$

$$\text{➤ } \frac{1}{13} = 7\frac{6}{13}\%$$

➤ $\frac{1}{14} = 7\frac{1}{7}\% = 7.14\%$

➤ $\frac{1}{15} = 6.66\% = 6\frac{2}{3}\%$

➤ $\frac{1}{16} = 6.25\% = 6\frac{1}{4}\%$

➤ $\frac{1}{17} = 5.88\%$

➤ $\frac{1}{18} = 5.55\%$

➤ $\frac{1}{19} = 5.26\%$

➤ $\frac{1}{20} = 5\%$

➤ Find the percent value of $\frac{5}{9}$

$$\frac{1}{9} = 11\frac{1}{9}\% = \left(11 + \frac{1}{9}\right)\%$$

$$\frac{5}{9} = 5 \left(11 + \frac{1}{9}\right)\% = 55\frac{5}{9}\%$$

➤ Find the percent value of $\frac{3}{8}$

$$\frac{1}{8} = 12\frac{1}{2}\% = \left(12 + \frac{1}{2}\right)\%$$

$$\frac{3}{8} = 3 \left(12 + \frac{1}{2}\right)\% = 37\frac{1}{2}\%$$

➤ Find the percent value of $\frac{5}{6}$

$$\frac{1}{6} = 16\frac{2}{3}\% = \left(16 + \frac{2}{3}\right)\%$$

$$\frac{5}{6} = 5 \left(16 + \frac{2}{3}\right)\% = 80 + 3\frac{1}{3}\% = 83\frac{1}{3}\%$$

➤ Find the percent value of $\frac{2}{3}$

$$\frac{1}{3} = 33\frac{1}{3}\% = \left(33 + \frac{1}{3}\right)\%$$

$$\frac{2}{3} = 2 \left(33 + \frac{1}{3}\right)\% = 66\frac{2}{3}\%$$

➤ Find the percent value of $\frac{5}{8}$

$$\frac{1}{8} = 12\frac{1}{2}\% = \left(12 + \frac{1}{2}\right)\%$$

$$\frac{5}{8} = 5 \left(12 + \frac{1}{2}\right)\% = 60 + \frac{5}{2} = 62\frac{1}{2}\%$$

- Find the percent value of $\frac{4}{7}$

$$\frac{1}{7} = 14\frac{2}{7}\% = \left(14 + \frac{2}{7}\right)\%$$

$$\frac{4}{7} = 4 \left(14 + \frac{2}{7}\right) = 57\frac{1}{7}\%$$

- Find the percent value of $\frac{7}{12}$

$$\frac{1}{12} = 8\frac{1}{3}\% = \left(8 + \frac{1}{3}\right)\%$$

$$\frac{7}{12} = 7 \left(8 + \frac{1}{3}\right) = 58\frac{1}{3}\%$$

TYPE – 3

How to change the fraction which % value more than 100%

- Find the percent value of $\frac{7}{5}$

$$\frac{7}{5} = \frac{5}{5} + \frac{2}{5}$$

$$= 100\% + 40\%$$

$$= 140\%$$

- Find the percent value of $\frac{33}{7}$

$$\frac{33}{7} = \frac{28}{7} + \frac{5}{7}$$

$$= 400\% + 71\frac{3}{7}\% = 471\frac{3}{7}\%$$

- Find the percent value of $\frac{23}{12}$

$$\frac{23}{12} = \frac{12}{12} + \frac{11}{12}$$

$$= 100\% + 91\frac{2}{3}\% = 191\frac{2}{3}\%$$

- Find the percent value of $\frac{41}{6}$

$$\frac{41}{6} = \frac{42}{6} - \frac{1}{6}$$

$$= 700\% - 16\frac{2}{3}\% = 683\frac{1}{3}\%$$

TYPE – 4

- Find the fraction value of $157\frac{1}{7}\%$

$$157\frac{1}{7}\% = 100\% + 57\frac{1}{7}\% \\ = 1 + \frac{4}{7} = \frac{11}{7}$$

- Find the fraction value of $616\frac{2}{3}\%$

$$616\frac{2}{3}\% = 600\% + 16\frac{2}{3}\% \\ = 6 + \frac{1}{6} = \frac{37}{6}$$

- Find the fraction value of $208\frac{1}{3}\%$

$$208\frac{1}{3}\% = 200\% + 8\frac{1}{3}\% \\ = 2 + \frac{1}{12} = \frac{25}{12}$$

1. 20% of 35% of a number is 175. What is 18% of that number?

- 1) 350
- 2) 450
- 3) 250
- 4) 300
- 5) None of these

Ans: 2

2. The sum of 60% of a number and 15% of the same number is 20,640. What is 55% of that number?

- 1) 15,136
- 2) 15,616
- 3) 15,316
- 4) 15,216
- 5) None of these

Ans: 3

3. $\frac{1}{2}$ of $\frac{3}{4}$ th of $\frac{2}{5}$ th of a number is 3621. What is 35% of that number?

- 1) 8049
- 2) 8849
- 3) 8989
- 4) 8449
- 5) None of these

Ans 4

4. One third of a first number is equal to twenty-four per cent of the second number. The second number is equal to one – sixth of the third number. The value of the third number is 4200. What is 50% of the first number?

- 1) 350
- 2) 452
- 3) 252
- 4) 250
- 5) None of these

Ans: 3

5. Mr. Verma's salary is increased from 39,000 to 65,000. Find the approx. increase %?

- 1) 67%
- 2) 50%
- 3) 25%
- 4) 33%
- 5) None of these

Ans: 1

6. The population of a village has increased from 32,000 to 40,000. Find the increase percent?

- 1) 35%

- 2) 45%
- 3) 25%
- 4) 30%
- 5) None of these

Ans: 3

7. Population of a town has decreased from 60,000 to 45,000. Find the decrease percent?

- 1) 35%
- 2) 45%
- 3) 25%
- 4) 30%
- 5) None of these

Ans: 3

8. Raghu's salary is 60% more than that of Vasu. How much % is Vasu's salary less than that of Raghu?

- 1) 35%
- 2) 45%
- 3) 37.5%
- 4) 30%
- 5) None of these

Ans: 3

9. Price of a shirt is 60% less than that of price of a trouser. Price of trouser is how much percent more than that of price of shirt?

- 1) 350%
- 2) 450%
- 3) 250%
- 4) 150%
- 5) None of these

Ans: 4

10. The price of a T.V. Set is decreased by 30% and hence the sale is increased by 30%.

What will be the effect on the total revenue of the shop?

- 1) 5% decrease
- 2) 4.5% increase
- 3) 9% decrease
- 4) 10% decrease
- 5) None of these

Ans: 3

Solution

Sol-1-

Let the number be x .

$\therefore 20\% \text{ of } 35\% \text{ of } x = 175$

$$\Rightarrow x = \frac{175 \times 100 \times 100}{20 \times 35}$$

$$\Rightarrow 18\% \text{ of } x = \frac{175 \times 100 \times 100}{20 \times 35} \times \frac{18}{100} = 450$$

Sol-2-

Let the number be x .

$\therefore 60\% \text{ of } x + 15\% \text{ of } x = 20,640$

$$\Rightarrow x = 20,640 \times \frac{100}{75}$$

$$\therefore 55\% \text{ of } x = 20,640 \times \frac{100}{75} \times \frac{55}{100} = 15,136$$

Sol-3-

Let the number be x .

$$\therefore \frac{1}{2} \text{ of } \frac{3}{4} \times \text{of } \frac{2}{5} \text{ of } x = 3,621$$

$$\Rightarrow x = 3621 \times \frac{2 \times 4 \times 5}{1 \times 3 \times 2}$$

$$\therefore 35\% \text{ of } x = 3,621 \times \frac{2 \times 4 \times 5}{1 \times 3 \times 2} \times \frac{35}{100}$$

$$\Rightarrow x = 8,449$$

Sol-4

Given: 3rd no. = 4,200

$$\therefore 2^{\text{nd}} \text{ no.} = \frac{1}{6} \text{ of } 4,200 = 700,$$

$$\Rightarrow \frac{1}{3} \text{ of } 1^{\text{st}} \text{ no.} = 24\% \text{ of } 700$$

$$\Rightarrow 1^{\text{st}} \text{ no.} = 24 \times 7 \times 3 = 504$$

$$\Rightarrow 50\% \text{ of } 504 = 252$$

Sol-5-

$$\text{Reqd. \% increase} = \frac{65000 - 39000}{39000} \times 100 = \frac{26000}{39000} \times 100 = 66.67\%$$

Sol-6-

$$\text{Reqd. \% increase} = \frac{40000 - 32000}{32000} \times 100 = \frac{8000}{32000} \times 100 = 25\%$$

Sol-7-

$$\text{Reqd. \% decrease} = \frac{60000 - 45000}{60000} \times 100 = \frac{15000}{60000} \times 100 = 25\%$$

Sol-8-

Let, Vasu's Salary = Rs.100

$\therefore$ Raghu's Salary = Rs.160

$$\therefore \text{Reqd. \%} = \frac{60}{160} \times 100 = 37.5 \%$$

Sol-9-

Let, the Price of Trouser = Rs.100

$\therefore$ The price of Shirt = Rs. 40

$$\therefore \text{Reqd. \%} = \frac{60}{40} \times 100 = 150 \%$$

Sol-10-

Let, the original price of T.V Set = Rs.100

$\therefore$ Now, price of the T.V Set = Rs. $100 \times 70\% \times 130\% = \text{Rs. } 91$

$\therefore$ Reqd. \% = $100 - 91 = 9 \%$ decrease